

Name: JAY ARRE TALOSIG

Subject & Section: CTBASNTL – COM231

Professor: Mr. Eliseo Q. Ramirez

Assignment #3

1. What is Dynamic and Static IP address?

- **Static IP Address:** A static IP address is a fixed address that does not change. It is manually assigned to a device and remains constant over time. Static IPs are often used for servers or devices that need to be consistently reachable, such as web servers, email servers, and network printers.
- **Dynamic IP Address:** A dynamic IP address is assigned by a DHCP (Dynamic Host Configuration Protocol) server and can change over time. When a device connects to a network, it is assigned an available IP address from a pool of addresses. Dynamic IPs are commonly used for personal devices like laptops and smartphones, as they allow for more efficient use of IP address space.

2. What are the different classes of IPv4 address?

IPv4 addresses are divided into five classes based on their leading bits and the range of addresses they cover:

- **Class A:**
 - Range: 0.0.0.0 to 127.255.255.255
 - Leading Bits: 0
 - Default Subnet Mask: 255.0.0.0
 - Used for large networks.
- **Class B:**
 - Range: 128.0.0.0 to 191.255.255.255
 - Leading Bits: 10
 - Default Subnet Mask: 255.255.0.0
 - Used for medium-sized networks.
- **Class C:**
 - Range: 192.0.0.0 to 223.255.255.255
 - Leading Bits: 110
 - Default Subnet Mask: 255.255.255.0
 - Used for small networks.

- **Class D:**
 - Range: 224.0.0.0 to 239.255.255.255
 - Leading Bits: 1110
 - Used for multicast groups.
- **Class E:**
 - Range: 240.0.0.0 to 255.255.255.255
 - Leading Bits: 1111
 - Reserved for experimental purposes.

3. What is ARP?

ARP (Address Resolution Protocol) is a network protocol used to find the hardware (MAC) address of a host from its IP address within a local area network (LAN). When a device wants to communicate with another device on the same network, it uses ARP to map the IP address to the corresponding MAC address. ARP broadcasts a request packet to all devices on the network, and the device with the matching IP address responds with its MAC address.

4. What is a port number in data communications, give 10 examples of port number and its use.

A **port number** is a numerical identifier in networking that is used to distinguish different services or applications running on a device. It allows multiple services to run on a single IP address by directing incoming traffic to the appropriate application.

Here are 10 examples of port numbers and their uses:

1. **Port 20:** FTP Data Transfer
2. **Port 21:** FTP Control (File Transfer Protocol)
3. **Port 22:** SSH (Secure Shell) for secure logins and file transfers
4. **Port 23:** Telnet for unencrypted text communications
5. **Port 25:** SMTP (Simple Mail Transfer Protocol) for email sending
6. **Port 53:** DNS (Domain Name System) for resolving domain names to IP addresses
7. **Port 80:** HTTP (Hypertext Transfer Protocol) for web traffic
8. **Port 443:** HTTPS (HTTP Secure) for secure web traffic
9. **Port 110:** POP3 (Post Office Protocol) for retrieving emails
10. **Port 143:** IMAP (Internet Message Access Protocol) for accessing emails on a mail server